Outline

- Central Processing Unit (CPU)
- Components of CPU
- Working of CPU
- Types of CPUs
- CPU Performance Factors


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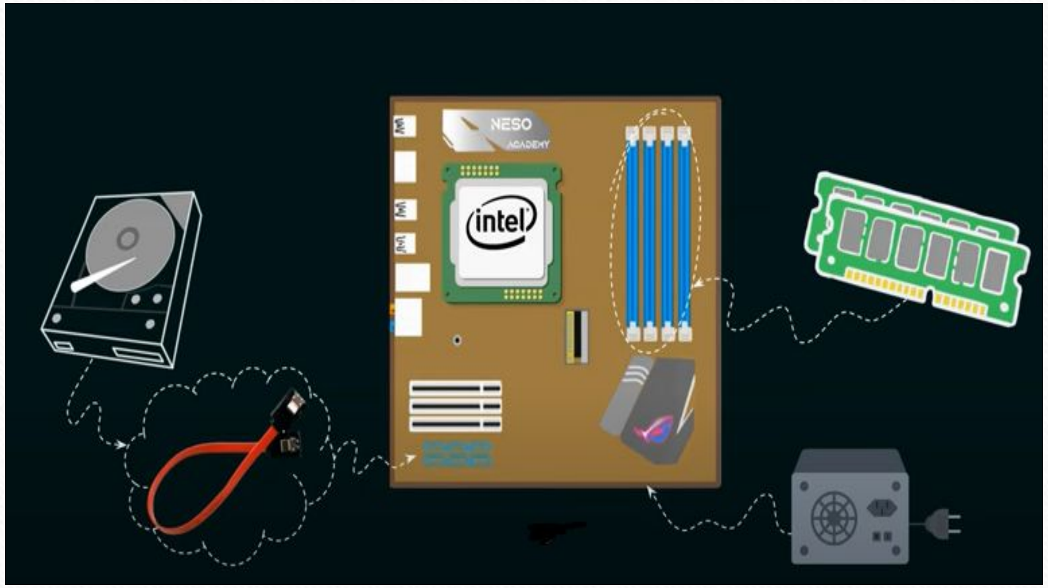
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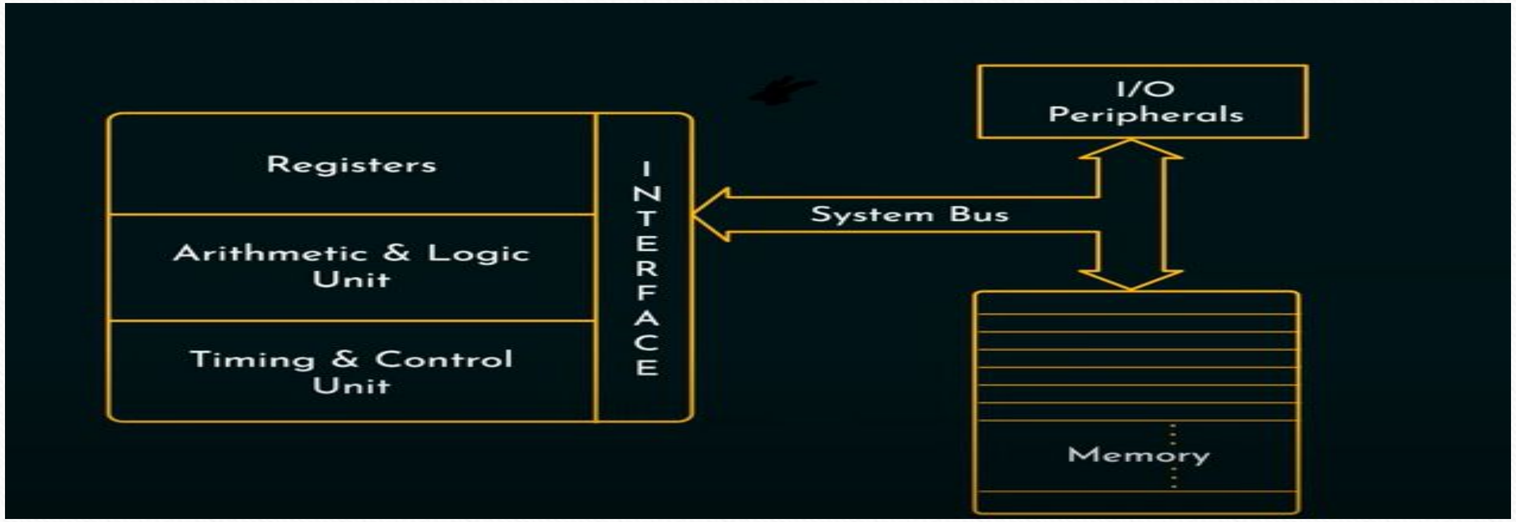
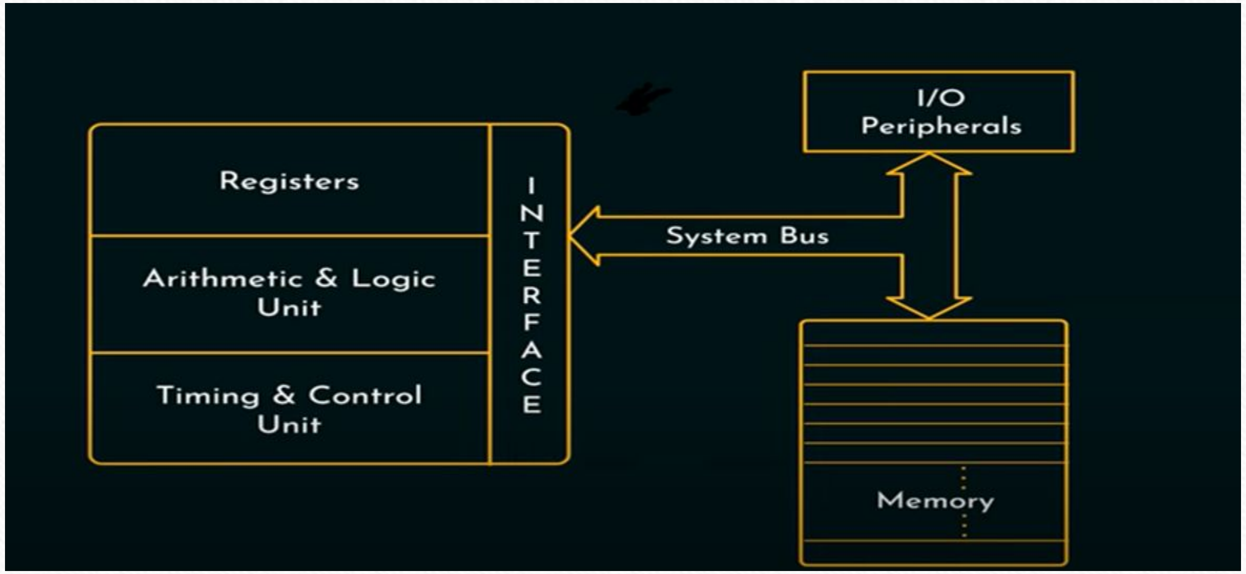

Registers
Arithmetic & Logic Unit
Timing & Control Unit
INTERFACE
System Bus
I/O Peripherals
Memory

Central Processing Unit (CPU)

- Every computer system has a unit whose primary purpose is to process data.
- This unit is the control center of the entire computer system.
- It accepts data from input devices, processes data and sends results to the printer or other output devices under control of a stored program.
- This unit is referred to as the microprocessor in a microcomputer and the central processing unit (CPU) in large computer systems.
- Both units perform basically the same functions.

Main Components of CPU



- ❖ Arithmetic Logic Unit (ALU)
- ❖ Control Unit
- ❖ Registers

Arithmetic/Logic Unit (ALU)

- ALU is the data processing unit of the microprocessor.
- Functions of ALU are :
  - Arithmetic operation (addition, subtraction, multiplication and division)
  - Logical operations (OR, AND, NOT etc..)
  - Decision Making
- ALU is the computer calculator.
- The ALU makes use of temporary storage areas referred to as registers.
- Data to be arithmetically manipulated are copied from memory and placed in registers for processing.

Control Unit (CU)

Function:

- Manages and coordinates the operations of the CPU.
- It fetches instructions from memory, decodes them, and directs other components to execute them.

Key Tasks:

- **Instruction Fetch**: Retrieves instructions from memory.
- **Instruction Decode:** Interprets the instructions.
- **Execution Control:** Sends signals to other components to perform operations.

Registers

Function: Small, fast storage locations within the CPU used to hold data, instructions, and addresses temporarily during processing.

Types:

- **General-Purpose Registers**: Store data and intermediate results.
- **Special-Purpose Registers**:
  - **Program Counter (PC)**: Holds the address of the next instruction to be executed.
  - **Instruction Register (IR)**: Holds the current instruction being executed.
  - **Memory Address Register (MAR)**: Stores the address of data to be accessed in memory.
  - **Memory Data Register (MDR)**: Temporarily holds data being transferred to or from memory.
  - **Status Register (Flag Register)**: Stores information about the state of the CPU (e.g., overflow, carry, zero flags).

Bus Interface Unit (BIU)

Function: Manages communication between the CPU and other components (e.g., memory, I/O devices) via buses.

Buses:

- **Data Bus**: Transfers data between CPU and memory.
- **Address Bus**: Carries memory addresses for data access.
- **Control Bus**: Transmits control signals (e.g., read/write).

Working of CPU (Fetch-Decode-Execute)

Step 1: Fetch (Retrieve the Instruction)

- The Control Unit (CU) fetches an instruction from RAM (main memory).
- The Program Counter (PC) holds the memory address of the next instruction to be executed.
- The instruction is loaded into the Instruction Register (IR).

Step 2: Decode (Interpret the Instruction)

- The Control Unit decodes the fetched instruction.
- It determines what the instruction is asking the CPU to do (e.g., mathematical operation, data transfer).
- If needed, the instruction is sent to the Arithmetic Logic Unit (ALU) for processing.

Step 3: Execute (Perform the Operation)

- The ALU executes arithmetic or logical operations if required.
- The result is stored in a **register** or sent back to **memory**.
- The **Program Counter (PC) is updated** to point to the next instruction.

Types of CPUs – Based on Core Count

- A **core** is an individual processing unit within a CPU that can execute instructions independently.
- CPUs are categorized based on the **number of cores**, affecting their **performance, multitasking ability, and power efficiency**

Single-Core CPU – Can execute only one instruction at a time.
Multi-Core CPU – Has multiple processing cores to handle parallel tasks.

✔ **Dual-Core** – 2 cores (e.g., Intel Core 2 Duo).
✔ **Quad-Core** – 4 cores (e.g., Intel Core i5).
✔ **Octa-Core** – 8 cores (e.g., AMD Ryzen 7).

Types of CPUs - Based on Instruction Set Architecture (ISA)

CISC (Complex Instruction Set Computing) :

- CISC processors have a complex set of instructions that can execute multiple tasks in a single instruction.
- This reduces the number of instructions required for a program to run, though the instructions themselves can take multiple clock cycles to execute.
- **Example:** Intel Core i7 (CISC-based x86) (Desktops and Laptops)

RISC (Reduce Instruction Set Computing) :

- RISC processors use a simplified set of instructions where each instruction is designed to execute in a single clock cycle.
- This allows faster processing but often requires more instructions to perform complex tasks.
- **Example :** Smartphones and Tablets

Difference between CISC and RISC

SL	CISC	RISC
1	Complex Instruction Set Computer	Reduced Instruction Set Computer
2	Large instruction set	Limited number of instructions
3	Number of registers are very small	Large number of registers are required
4	Pipelining is difficult	Pipelining is easy
5	Many addressing modes	Few addressing modes
6	Instruction take more than one cycle	An instruction execute in single clock cycle
7	Examples of CISC: VAX, Motorola 68000 family, System/360, AMD and the Intel x86 CPUs.	Example of RISC: ARM, PA-RISC, Power Architecture, Alpha, AVR, ARC and the SPARC.

Thank You